



University of Engineering & Management, Kolkata
University of Engineering & Management, Jaipur
Institute of Engineering & Management, Kolkata
B.Tech in CSE, IT & Allied Branches

DETAILED SYLLABUS

Course Code- PCCCS503

Course Title –Operating Systems

Credit – 3

Category – Professional Core Course

Semester – V

L:T:P:S – 3:0:0:0

Pre-requisite – Basic knowledge of Data Structures and Computer Organization.

Course Outcomes:

CO1	Explain the role, functionality, layering, abstractions, interfaces and major components of an operating system.
CO2	Describe process abstraction, system calls, process execution, scheduling and process-management mechanisms.
CO3	Analyse memory-management mechanisms, concurrency problems and synchronization-based solutions.
CO4	Explain file-system abstractions, file-system interfaces, persistent storage and file-system design principles.

Study Material	Coursera	NPTEL	LinkedInLearning	Infosys Springboard
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Module No.	Topic	Sub-topics	Mapping with Industry and International Academia	Lecture Hours	Textbook Mapping
1	Introduction to Operating Systems	Application requirements; systems stack and role of the operating system; resources, abstractions and interfaces; overview of operating-system components; examples and characteristics of different types of operating systems such as real-time, desktop and mobile operating systems; operating systems and operating-system distributions.	International Academia: https://online.stanford.edu/courses/cs111-operating-systems-principles AICTE-prescribed syllabus: AICTE Model Curriculum for UG Degree Course in CSE (PCC CS-403) Industry Mapping Linux/UNIX, Android, real-time operating systems.	2	Text Book 1: Ch. 2; Text Book 2: Ch. 1-2; Text Book 3: Ch. 1
2	Computer Organization and Architecture Refresher	Basic organization of hardware components; role of the operating system relative to hardware functionality, with examples related to the von Neumann architecture.	International Academia: https://online.stanford.edu/courses/cs111-operating-systems-principles AICTE-prescribed syllabus: AICTE Model Curriculum for UG Degree Course in CSE (PCC CS-403) Industry Mapping CPU privilege modes, interrupts and system-call interface.	1	Text Book 2: Ch. 1; Text Book 3: Ch. 1
3	Processes	Process abstraction: program versus process; Process Control Block (PCB); design of system calls, their invocation and basic operating-system handling; process-control system calls such as fork, wait, exec, getpid, getppid and variants; limited direct execution model.	International Academia: https://online.stanford.edu/courses/cs111-operating-systems-principles AICTE-prescribed syllabus: AICTE Model Curriculum for UG Degree Course in CSE (PCC CS-403) Industry Mapping POSIX process API, Linux process model.	5	Text Book 1: Ch. 4-6; Text Book 2: Ch. 6-7; Text Book 3: Ch. 7-8
4	Memory Management	Address bus and memory access; memory view of a process: heap, stack,	International Academia: https://online.stanford.edu/courses/cs111-operating-systems-principles	9	Text Book 1: Ch. 13-

Module No.	Topic	Sub-topics	Mapping with Industry and International Academia	Lecture Hours	Textbook Mapping
		code and data; process memory requirements; address-space abstraction using virtual memory and related system calls; address translation mechanisms: static mapping, segmentation and paging; page faults, page sharing, read/write permissions, swapping, process versus OS memory; motivation and mechanisms for memory bookkeeping and management; case studies of malloc and the role of the OS in transforming a program into a process.	<u>operating-systems-principles</u> AICTE-prescribed syllabus: <u>AICTE Model Curriculum for UG Degree Course in CSE (PCC CS-403)</u> Industry Mapping <i>virtual memory, paging, memory protection, Linux process address space.</i>		20; Text Book 2: Ch. 9
5	Process Management	Process life cycle from source code to execution; operating-system mode of execution; limited direct execution, interrupts and system calls; process context-switch mechanism and PCB state; scheduling policies, scheduling metrics and goals; examples involving interactive, real-time and priority scheduling.	International Academia: <u>https://online.stanford.edu/courses/cs111-operating-systems-principles</u> AICTE-prescribed syllabus: <u>AICTE Model Curriculum for UG Degree Course in CSE (PCC CS-403)</u> Industry Mapping <i>Linux schedulers, process priorities and context switching.</i>	4	Text Book 1: Ch. 7-8; Text Book 2: Ch. 8
6	Concurrency and Synchronization	Motivation and application, process and OS use cases; introduction to threads and the pthread API; synchronization primitives; limitations of software-only solutions; atomic instructions, test-and-set, spinlocks, mutexes, condition variables and semaphores; pthread synchronization API; case studies: producer-consumer, reader-writers and barriers; issues with concurrency	International Academia: <u>https://online.stanford.edu/courses/cs111-operating-systems-principles</u> AICTE-prescribed syllabus: <u>AICTE Model Curriculum for UG Degree Course in CSE (PCC CS-403)</u> Industry Mapping <i>POSIX threads, mutexes, semaphores and</i>	9	Text Book 1: Ch. 26-32; Text Book 3: Ch. 11-12

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		including race conditions, deadlocks and order violations.	<i>concurrent systems.</i>		
7	File Systems	Persistence and the file abstraction; hardware view of hard-disk architecture and interfacing; process view and system calls for file handling; roles and responsibilities of a file system; file-system design details: file and file-system metadata, directory structure and caching optimizations; file-system case study such as the UNIX file system.	International Academia: https://online.stanford.edu/courses/cs111-operating-systems-principles AICTE-prescribed syllabus: AICTE Model Curriculum for UG Degree Course in CSE (PCC CS-403) Industry Mapping <i>POSIX file API, UNIX/Linux file systems.</i>	6	Text Book 1: Ch. 36-40; Text Book 2: Ch. 4-5; Text Book 3: Ch. 3

Textbooks:

1. Operating Systems: Three Easy Pieces, Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau, Arpaci-Dusseau Books, LLC.
2. Operating System Concepts, Abraham Silberschatz, Peter B. Galvin and Greg Gagne, Wiley.
3. Operating Systems: Internals and Design Principles, William Stallings, Pearson.

Reference books / Online Resources:

1. Modern Operating Systems, Andrew S. Tanenbaum and Herbert Bos, Pearson.
2. Operating Systems: Principles and Practice, Thomas Anderson and Michael Dahlin, Recursive Books.
3. AICTE Model Curriculum for UG Degree Course in Computer Science and Engineering (2022), PCC CS-403 Operating Systems.
4. Stanford CS111: Operating Systems Principles; MIT 6.S081: Operating System Engineering.

CO-PO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	1	1							
CO2	3	3	2	2	1							
CO3	3	3	2	2	1							
CO4	3	2	2	1	2							